

1.7 Classical Mechanics as a Commutative C^* -algebra

The standard description of a physical system establishes geometry as the primary structure by defining the coordinate space before the algebra of continuous functions. The Gelfand theory of abelian C^* -algebras introduces a deep change of perspective for Hamiltonian mechanics. By exploiting the Gelfand-Naimark representation theorem one can completely reverse this relation and construct the theory starting from the abstract algebra of observables. The underlying compact Hausdorff topological space is then mathematically reconstructed as the intrinsic Gelfand spectrum of the algebra. This paradigm establishes that the algebraic encoding of physical quantities comes first while the geometrical description emerges consequently.

Basic topics in C^* -algebras theory

To reconstruct classical theory using only the algebra of observables, we first need to present C^* -algebras. This abstract mathematical structure recurs in both classical and quantum descriptions of observables. As explained in Brattelli and Robinson [3], this theory has profound applications in broad areas of mathematical physics. Here, we will present only the basic definitions to permit the reader to understand why Hamiltonian mechanics is the theory of abelian C^* -algebras.

We begin by constructing the definition of a $*$ -algebra.

Definition 1.7.1 (Algebra). *An algebra $\mathfrak{A}$ is a $\mathbb{C}$ -vector space equipped with an associative and distributive multiplication law which is compatible with the other vector space operations.*

Definition 1.7.2 ($*$ -algebra). *Let $\mathfrak{A}$ be an algebra as in Definition 1.7.1. An involution on $\mathfrak{A}$ is a map associating an element $A \mapsto A^*$ satisfying three distinct conditions:*

- $A^{**} = A$,
- $(AB)^* = B^*A^*$,
- $(\alpha A + \beta B)^* = \bar{\alpha}A^* + \bar{\beta}B^*$.

An algebra equipped with an involution map is called a $$ -algebra.*

Within a $*$ -algebra it is natural to isolate subsets that remain stable under the involution, since such subsets will later correspond to physically meaningful collections of observables. We therefore introduce the following notion of a selfadjoint subset.

Definition 1.7.3 (Selfadjoint subset). *Let $\mathfrak{A}$ be a $*$ -algebra in the sense of Definition 1.7.2. A subset $\mathfrak{B}$ of $\mathfrak{A}$ is called selfadjoint if $A \in \mathfrak{B}$ implies that $A^* \in \mathfrak{B}$.*

Some particular elements of this subset are the adjoint of themselves, those are called selfadjoint elements.

Definition 1.7.4 (Selfadjoint element). *Let $\mathfrak{A}$ be a $*$ -algebra in the sense of Definition 1.7.2. An element $A \in \mathfrak{A}$ is selfadjoint if $A = A^*$.*

We are now ready to endow an algebra with an operation that works as a measure of length of their element.

Definition 1.7.5 (Normed algebra). *An algebra $\mathfrak{A}$, defined in Definition 1.7.1, is a normed algebra if to each element $A \in \mathfrak{A}$ there is an associated real number $\|A\| \in \mathbb{R}$, called the norm of A , satisfying the requirements:*

- $\|A\| \geq 0$ and $\|A\| = 0$ if, and only if, $A = 0$,
- $\|\alpha A\| = |\alpha| \|A\|$,
- $\|A + B\| \leq \|A\| + \|B\|$ the triangle inequality,
- $\|AB\| \leq \|A\| \|B\|$ the product inequality.

Normed algebras are fundamental objects inside our construction, since the norm endows each of them with a notion of a particular topology called *uniform topology*.

Definition 1.7.6 (Uniform topology). *Let $\mathfrak{A}$ be a normed algebra as in Definition 1.7.5. The norm defines a metric topology on $\mathfrak{A}$ which is referred to as the uniform topology. The neighborhoods of an element $A \in \mathfrak{A}$ in this topology are given by*

$$\mathcal{U}_{A,\varepsilon} = \{B \in \mathfrak{A} \mid \|B - A\| < \varepsilon\},$$

where $\varepsilon \in \mathbb{R}^+$.

Having defined the norm, we can now endow a vector space with that operation and consequently generate an algebra. A vector space equipped with a norm is called a *normed vector space* and adding completeness makes a normed vector space a Banach space.

Definition 1.7.7 (Banach Space on $\mathbb{C}$). *A Banach space is a normed $\mathbb{C}$ -vector space $(X, \|\cdot\|)$ such that the metric induced by the norm, defined by*

$$d(x, y) := \|x - y\|, \quad x, y \in X,$$

is complete.

Equipping a Banach space with a compatible associative multiplication yields the notion of a Banach algebra. This combination is what will later allow observables to be manipulated algebraically while still controlling limiting procedures such as sequences of approximating observables.

Definition 1.7.8 (Banach algebra). *Let $\mathfrak{A}$ be a normed algebra as in Definition 1.7.5. If $\mathfrak{A}$ is complete with respect to the uniform topology, as in Definition 1.7.6, it is called a Banach algebra.*

A Banach algebra alone still lacks any notion of adjunction between its elements, a feature that will be needed to encode the passage from an observable to its complex conjugate at the algebraic level. So, we can endow a Banach algebra with an involution as following.

Definition 1.7.9 (Banach $*$ -algebra). *A Banach $*$ -algebra is a complete normed algebra $\mathfrak{A}$ as in Definition 1.7.5 equipped with an involution, as in Definition 1.7.2, such that the norm satisfies the isometric property*

$$\|A\| = \|A^*\|, \quad \forall A \in \mathfrak{A}.$$

Completeness assures that every Cauchy sequence in the space converges to an element thereof, thus guaranteeing that approximations of observables are well-defined. In a Banach $*$ -algebra, the norm and involution are only constrained by $\|A\| = \|A^*\|$. Imposing the stronger identity $\|A^*A\| = \|A\|^2$ makes the norm compatible with the algebraic structure and defines a C^* -algebra.

Definition 1.7.10 (C^* -algebra). *A C^* -algebra is a Banach $*$ -algebra $\mathfrak{A}$, as in Definition 1.7.9 with the property $\|A^*A\| = \|A\|^2 \forall A \in \mathfrak{A}$.*

C^* -algebras constitute a fundamental framework in mathematical physics where algebraic structures are canonically endowed with a complete metric topology. The abstraction of bounded operators leads to the systematic study of Banach algebras and their involutions. This rigorous topological approach allows the characterization of states and observables in both classical and quantum theories. The most important classification of C^* -algebras is represented by those in which there is an element behaving exactly like a unit. Those algebras are called *unital*.

Definition 1.7.11 (Unital C^* -algebra). *A unital C^* -algebra is a C^* -algebra $\mathfrak{A}$, as in Definition 1.7.10, containing an identity element $\mathbb{1}$ such that $A = \mathbb{1}A = A\mathbb{1}$ for all $A \in \mathfrak{A}$.*

The identity element introduced above plays the role of a distinguished observable acting trivially on every other one. Before relying on this notion, it is worth verifying that, when it exists, it is uniquely determined by the algebra.

Proposition 1.7.12. *A C^* -algebra $\mathfrak{A}$ can possess at most one identity element.*

Proof. If a second identity element $\mathbb{1}'$ exists it must satisfy $\mathbb{1}' = \mathbb{1}\mathbb{1}' = \mathbb{1}$. □

C*-algebra description of Hamiltonian Mechanics

We now apply the abstract framework developed above to the specific case of Hamiltonian mechanics. In the classical setting, physical intuition suggests starting from the phase space and only afterward considering functions defined on it. Our goal will be to indicate that the reverse order is equally legitimate and, in fact, more fundamental. We now carry out this reconstruction explicitly, showing how the familiar objects of classical mechanics, observables, states, and dynamics, all arise once the phase space is replaced by its associated commutative C*-algebra.

Throughout this section, we will refer to X as a compact Hausdorff topological space.

We begin by algebraically characterizing the observables on a generic compact phase space Γ , in the sense of Definition ???. Let $\mathcal{P}$ be the algebra of *polynomial observables* generated by the phase space coordinates q and p . Those represent the act of measuring position or momentum of a physical system. These coordinates have different values for different points on Γ , namely, they separate points. $\mathcal{P}$ also includes constant functions. Consequently it fulfills the hypotheses of the Stone-Weierstrass Theorem [2, Theorem IV.9]. This fundamental result establishes that $\mathcal{P}$ is dense within $C^0(\Gamma; \mathbb{R})$ under the uniform norm topology that, in this space, is defined analytically as:

$$\|f\|_\infty = \sup_{x \in \Gamma} |f(x)| \quad \forall f \in \mathcal{P}.$$

Taking the closure with respect to this norm exactly recovers the full space $\overline{\mathcal{P}}^{\|\cdot\|_\infty} = C^0(\Gamma; \mathbb{R})$.

Since the subsequent algebraic machinery relies on Banach algebras over $\mathbb{C}$, we must complexify the space to the C*-algebra $C^0(\Gamma; \mathbb{C})$. The original real observables are then recovered as the elements invariant under complex conjugation, a procedure that motivates the general axiomatic framework.

Definition 1.7.13 (Classical Gelfand Algebra). *Let X be a compact Hausdorff space. The classical Gelfand algebra is the commutative unital C*-algebra, as in Definition 1.7.11, $\mathfrak{A} = C^0(X)$, equipped with:*

1. *pointwise multiplication* $(fh)(x) = f(x)h(x)$,
2. *the identity element* $\mathbb{1}(x) = 1$,
3. *the involution* $f^*(x) = \overline{f(x)}$,
4. *the supremum norm* $\|f\|_\infty = \sup_{x \in X} |f(x)|$.

In the present case $X = \Gamma$, it descends that $\mathfrak{A} = C^0(\Gamma)$ and the C*-algebra condition follows from $\|f^*f\|_\infty = \|f\|_\infty^2$.

Inside this algebra there are elements yielding a complex valued when evaluated. This is problematic for the physical interpretation of the quantity associated with those elements. Because of this, we will refer to *observables* as the selfadjoint elements of the classical Gelfand algebra presented in Definition 1.7.13.

Definition 1.7.14 (Classical Observable Algebra). *Let $\mathfrak{A}$ be a classical Gelfand algebra, as in Definition 1.7.13. The classical observable algebra $\mathfrak{A}_{sa}$ is the subset of selfadjoint elements of $\mathfrak{A}$ as seen in Definition 1.7.4.*

A more detailed discussion about the physical quantities really measurable will be done at the end of this chapter.

Having analysed the algebraic nature of observables, we now turn to states. In an operational point of view, if we prepare the system to be in a particular state ω and we measure an observable $f \in \mathfrak{A}_{sa}$ n times, we can define its mean value with

$$\langle f \rangle_n^{(\omega)} := \frac{1}{n} \sum_{i=1}^n m_i^{(\omega)}(f),$$

where $m_i^{(\omega)}(f)$ represent the i -th measurement. The limit of this expression, assuming that it converges experimentally, can be defined dually as

$$\omega(f) := \lim_{n \rightarrow \infty} \langle f \rangle_n^{(\omega)} \quad \forall f \in \mathfrak{A}_{sa}$$

In this picture, a state ω acts on observables, yielding the average of the measurements. However, we can extend this definition to a generic unital C^* -algebra $\mathfrak{A}$, presented in Definition 1.7.11, without attaching a physical interpretation. This functional must be linear and must preserve the positivity of observables. It descends that $\omega(f^*f) \geq 0$ for all $f \in \mathfrak{A}$. It can be also shown that positivity implies the Cauchy-Schwartz Inequality, namely

$$|\omega(A^*B)| \leq (\omega(A^*A) \omega(B^*B))^{1/2} \quad \forall A, B \in \mathfrak{A}.$$

Therefore, if $\mathfrak{A}$ is unital, we can impose $B = \mathbb{1}$ and square both members. By positivity we obtain that $\omega(\mathbb{1}) > 0$, if ω is non-trivial. This proves that one can always normalize a state, unless $\omega = 0$, through

$$\omega \mapsto \omega_{norm} = \omega(\mathbb{1})^{-1} \omega.$$

We can formalize this concept in the following definition.

Definition 1.7.15 (State on a C^* -Algebra). *Let $\mathfrak{A}$ be a unital C^* -algebra as in Definition 1.7.11. A state on $\mathfrak{A}$ is a linear functional $\omega : \mathfrak{A} \rightarrow \mathbb{C}$ such that*

1. $\omega(A^*A) \geq 0$ for all $A \in \mathfrak{A}$ *positivity,*
2. $\omega(\mathbb{1}) = 1$ *normalisation.*

The set of all states is denoted by $\mathcal{S}(\mathfrak{A})$.

States on a C^* -algebra are not isolated objects, since a convex combination of two physically admissible preparations of a system is itself a physically admissible preparation. Thus, we can analyse the properties of this space.

Proposition 1.7.16 (Convexity of the State Space). *Let $\mathfrak{A}$ be a unital C^* -algebra as in Definition 1.7.11. The set $\mathcal{S}(\mathfrak{A})$ of all states on $\mathfrak{A}$ is convex.*

Proof. Let $\omega_1, \omega_2 \in \mathcal{S}(\mathfrak{A})$ and let $t \in [0, 1]$. Define $\omega = t\omega_1 + (1-t)\omega_2$. Linearity is immediate. Positivity follows because for any $A \in \mathfrak{A}$,

$$\omega(A^*A) = t\omega_1(A^*A) + (1-t)\omega_2(A^*A) \geq 0.$$

Normalisation holds since $\omega(\mathbb{1}) = t\omega_1(\mathbb{1}) + (1-t)\omega_2(\mathbb{1}) = t + (1-t) = 1$. Thus $\omega \in \mathcal{S}(\mathfrak{A})$. Hence $\mathcal{S}(\mathfrak{A})$ is convex. $\square$

Given X , a compact Hausdorff space, a state ω on the unital C^* -algebra $C^0(X)$ is automatically continuous and therefore by the Riesz-Markov Representation Theorem [2, Theorem IV.14] it defines a unique Borel measure μ_ω on X such that

$$\omega(f) = \int_X f d\mu_\omega, \quad \mu_\omega(X) = \omega(\mathbb{1}) = 1. \quad (1.1)$$

Thus, if we restrict a state to act only on the classical observable algebra defined in Definition 1.7.14, then we can interpret it as probability distribution on the phase space, and observables become random variables. This probabilistic interpretation is essential in statistical mechanics.

In this description, classical mechanics is recovered as a special case. In fact, we can take μ_{p_0} as the probability measure concentrated on the point $P_0 = \{q_0, p_0\}$ of a compact phase space Γ , in the sense of Definition ???. Then μ_{p_0} is a Dirac δ function and the corresponding state ω_{p_0} is

$$\omega_{p_0}(f) = \int_\Gamma f d\mu_{p_0} = f(P_0). \quad (1.2)$$

Such states are also called *pure states* since they cannot be written as convex non-trivial linear combinations of other states. All the measurements yield the same result, $f_\omega = \omega(f)$, so there is no dispersion and probabilistically speaking the variance of that state vanishes. We can formalize these concepts generally in the following way.

Definition 1.7.17 (Pure State). A state $\omega \in \mathcal{S}(\mathfrak{A})$, as in Definition 1.7.15, is called pure if it is an extreme point of the convex set $\mathcal{S}(\mathfrak{A})$. That is, whenever $\omega = t\omega_1 + (1-t)\omega_2$ with $0 < t < 1$ and $\omega_1, \omega_2 \in \mathcal{S}(\mathfrak{A})$, then $\omega_1 = \omega_2 = \omega$.

Now, we extend the statistical definition of variance to our algebraic framework. This notion will provide the precise sense in which pure states correspond to dispersion-free measurements, independently of their characterization as extreme points of the state space.

Definition 1.7.18 (Variance of a State). Let $\mathfrak{A}$ be a unital C^* -algebra, as in Definition 1.7.11, and let $\omega \in \mathcal{S}(\mathfrak{A})$ be a state, defined in Definition 1.7.15. The variance of an element $A \in \mathfrak{A}$ in the state ω is defined as

$$\text{Var}_\omega(A) := \omega(A^*A) - \omega(A^*)\omega(A).$$

The quantity is non-negative by the Cauchy–Schwarz inequality applied to the positive linear functional ω . In particular, since the dispersion-free character of their measurements, pure states show a strong form of this property.

Proposition 1.7.19 (Zero Variance for Pure States). Let $\mathfrak{A}$ be a commutative unital C^* -algebra as in Definition 1.7.11 and let $\omega \in \mathcal{S}(\mathfrak{A})$ be a pure state, shown in Definition 1.7.17. Then

$$\text{Var}_\omega(A) = 0 \quad \forall A \in \mathfrak{A}.$$

Proof. By a standard result for commutative C^* -algebras, every pure state is multiplicative, see, e.g., [3, Proposition 2.3.27]. Hence $\omega(AB) = \omega(A)\omega(B)$ for all $A, B \in \mathfrak{A}$. In particular, $\omega(A^*A) = \omega(A^*)\omega(A)$. Substituting into Definition 1.7.18 yields $\text{Var}_\omega(A) = 0$ for every $A \in \mathfrak{A}$. $\square$

Later, in Proposition 1.7.22, this proposition will be extended to an equivalent definition of pure state.

To conclude the excursus on observables and states, we establish a fundamental duality between these two notions. On one hand, given $\mathfrak{A}$ a unital commutative C^* -algebra, every state $\omega \in \mathcal{S}(\mathfrak{A})$ induces an evaluation functional which, when restricted to the classical observable algebra $\mathfrak{A}_{sa}$ introduced in Definition 1.7.14, yields strictly real values. This property is mathematically guaranteed by the positivity of the state, ensuring that $f = f^*$ implies $\omega(f) \in \mathbb{R}$. Crucially, because every element $A \in \mathfrak{A}$ admits a unique decomposition into selfadjoint components as $A = A_1 + iA_2$, with $A_1, A_2 \in \mathfrak{A}_{sa}$, the action of a state on the entire complex C^* -algebra is completely and uniquely determined by its values on the physical observables.

Conversely, the uniqueness of a state given its observable values relies on the topological properties of the Gelfand spectrum X . Since X is compact and Hausdorff, it satisfies the hypothesis of Urysohn's lemma [2, Theorem IV.7]. It states that for any two distinct points P and Q in X , there exists a strictly real-valued function $f \in C^0(X; \mathbb{R}) \cong \mathfrak{A}_{sa}$ such that $f(P) = 0$ and $f(Q) = 1$. In terms of pure states, this yields $\omega_P(f) = 0$ and $\omega_Q(f) = 1$. Thus, the classical observable algebra $\mathfrak{A}_{sa}$ separates points on X . If $\omega_P(f) = \omega_Q(f)$ for every observable $f \in \mathfrak{A}_{sa}$, then necessarily $P = Q$.

Combining these algebraic and topological findings yields a fundamental duality relation: a physical state is completely characterized by the collection of its values on the selfadjoint observables, and the algebra of such observables is structurally sufficient to reconstruct the points of the underlying state space.

Having established observables and states, we now investigate dynamics on a compact phase space Γ as in Definition ???. Time evolution of the system $\{q, p\} \rightarrow \{q_t, p_t\}$ is continuous in time $t \in \mathbb{R}$, with a continuous dependence on the initial data at time t_0 and therefore it defines a one-parameter family of continuous invertible maps $\alpha_{t_0, t}$ of $C^0(\Gamma; \mathbb{R})$ into itself. These maps preserve all the algebraic relations, including the $*$ -operation. Such maps are called $*$ -automorphisms. Thus, the dynamics is encoded by a one-parameter group of automorphisms of the observable algebra.

Definition 1.7.20 (Time Evolution). Let $\mathfrak{A}$ be a unital C^* -algebra as in Definition 1.7.11. A time evolution is a strongly continuous one-parameter group of $*$ -automorphisms $\alpha_t : \mathfrak{A} \rightarrow \mathfrak{A}$, $t \in \mathbb{R}$, such that:

1. $\alpha_0 = \text{id}$,
2. $\alpha_{s+t} = \alpha_s \circ \alpha_t$ for all $s, t \in \mathbb{R}$,

3. each α_t is linear, multiplicative, preserves the involution, and is norm-continuous,
4. for every $A \in \mathfrak{A}$, the map $t \mapsto \alpha_t(A)$ is continuous in the norm topology.

By duality, the time evolution of states is given by $(\alpha_t^* \omega)(A) = \omega(\alpha_t(A))$ for all $A \in \mathfrak{A}$. In addition, since α_t are $*$ -automorphisms, they automatically preserve every selfadjoint sector $\mathfrak{A}_{sa}$, ensuring that real physical observables evolve into real physical observables.

Consequently, we can state the conclusion that: *A classical system is completely defined by the pair $(\mathfrak{A}, \alpha_t)$ where $\mathfrak{A}$ is a classical Gelfand algebra as in Definition 1.7.13 and α_t is a time evolution on it, as in Definition 1.7.20. To fix the initial conditions, it must also be given a state $\omega \in \mathcal{S}(\mathfrak{A})$, as in Definition 1.7.15.*

Abelian C^* -algebras and Gelfand's Theorem

The duality between states and observables is not yet fully explicit, because we have assumed the space X rather than derived it from the algebra. We now fill this gap by reconstructing X directly from $\mathfrak{A}$, using a special class of linear maps called characters. Since these characters turn out to be the pure states we have already discussed, this construction provides the missing algebraic definition to reconstruct the phase space. It thereby completes the reversal of perspective announced at the beginning of the chapter.

Definition 1.7.21 (Character of an Abelian C^* -Algebra). *Let $\mathfrak{A}$ be an abelian C^* -algebra as in Definition 1.7.10. A character ω of $\mathfrak{A}$ is a nonzero linear map, $\omega : \mathfrak{A} \rightarrow \mathbb{C}$ such that*

$$\omega(AB) = \omega(A)\omega(B)$$

for all $A, B \in \mathfrak{A}$. The Gelfand spectrum or just spectrum $\sigma(\mathfrak{A})$ is the set of all characters on $\mathfrak{A}$.

By now, characters are only an abstract definition but, in fact, those are exactly what we have called earlier pure states. To establish this, we need the following result.

Proposition 1.7.22 (Equivalence of Pure States and Characters). *Let ω be a nonzero linear functional over the abelian C^* -algebra $\mathfrak{A}$, seen in Definition 1.7.10. The following conditions are equivalent*

1. ω is a pure state
2. ω is a character

Hence the spectrum $\sigma(\mathfrak{A}) \subseteq \mathfrak{A}^$.*

See [3, Proposition 2.3.27] for a complete proof of this equivalence. Having introduced characters as an equivalent way to identify pure states on a C^* -algebra, we now address a more abstract matter. The uniform topology considered before in Definition 1.7.6 is excessively fine for the character space. Consequently, we seek a formalisation of pointwise convergence, coarser than the uniform topology and more practical.

Definition 1.7.23 (Weak and Weak* Topologies). *Let Ω be a Banach space as in Definition 1.7.7 and let Ω^* be its topological dual space. The weak topology on X is the coarsest topology such that every linear functional f in Ω^* is continuous.*

Analogously, the weak topology on Ω^* is the coarsest topology such that for every element $x \in \Omega$ the evaluation map $f : \Omega^* \rightarrow \mathbb{C}$ such that $f \mapsto f(x)$ is continuous.*

Remark 1.7.24. Weak* topology, stated in Definition 1.7.23, induces a notion of convergence in Ω^* . A sequence f_α in Ω^* converges to a functional f in the weak* topology if and only if $f_\alpha(x) \rightarrow f(x) \forall x \in \Omega$.

The primary motivation for introducing these topologies lies in their compactness properties. The standard norm topology on an infinite dimensional space is rigorously too fine to allow bounded sets to be compact and leads to unexpected results. By coarsening the topology of $\mathfrak{A}^*$ to only control pointwise convergence, the Banach-Alaoglu theorem [2, Theorem IV.21] guarantees that a closed unit ball of the dual space $\mathfrak{A}^*$ is strictly compact under the weak* topology. This feature ensures that the state space of a C^* -algebra is compact allowing the application of the representations theorems and guaranteeing the existence of states in thermodynamic limits.

The compactness properties of the weak topology are essential for the Gelfand-Naimark representation theorem which is the central result of this section.

Theorem 1.1 (Gelfand-Naimark representation theorem). *Let $\mathfrak{A}$ be an abelian C^* -algebra and X be the spectrum of $\mathfrak{A}$, as in Definition 1.7.21, equipped with the weak* topology induced by $\mathfrak{A}^*$. Then $\mathfrak{A}$ is isomorphic to the algebra $C_0(X)$ of continuous functions over X which vanish at infinity.*

A complete proof of this theorem is provided in [3, Theorem 2.1.11]. As a special case we state this corollary that is also proved in [3, Theorem 2.1.11B].

Corollary 1.7.25. *It follows from Theorem 1.1 that X is a locally compact Hausdorff space which is compact if, and only if, $\mathfrak{A}$ is unital, as in Definition 1.7.11. This makes $\mathfrak{A} \cong C^0(X)$ the space of continuous functions over X .*

The Gelfand-Naimark Theorem 1.1 establishes a duality between algebraic structures and topology, serving as the fundamental bridge to reconstruct classical mechanics starting from an arbitrary commutative C^* -algebra. This framework allows the extraction of a locally compact Hausdorff space X , known as the Gelfand spectrum, which is topologically formed by the characters of the algebra. By virtue of Proposition 1.7.22, these characters coincide exactly with the pure states of the system.

The topological nature of X is intimately tied to the algebraic properties. As we have seen earlier, if the algebra is unital, the closed unit ball of $\mathfrak{A}^*$ is compact in the weak* topology. The presence of the identity element forces the spectrum to be closed and bounded away from the trivial zero functional, rendering X strictly compact. Consequently, the algebraic isomorphism uniquely maps the abstract algebra to the space $C_0(X)$ of continuous functions vanishing at infinity, which naturally reduces to the full space $C^0(X)$ in the compact, unital case.

Reconstruction of the Symplectic Manifold

The Gelfand isomorphism identifies the topological spectrum X purely as a compact Hausdorff space, discarding all differentiable and symplectic information. To recover the phase space of classical mechanics, the C^* -algebraic framework must be augmented with additional algebraic structures. These structures are precisely captured by the two following definitions. The Poisson algebra formalizes the Lie bracket of observables, while the smooth dense subalgebra supplies the required differentiability. The compatibility between them, enforced by the Leibniz rule, ensures that the resulting geometry is coherent. Crucially, the choice of such a subalgebra is not inherent to $\mathfrak{A}$ itself and constitutes the central external input of the reconstruction procedure.

We begin by defining the main structure that permits to recover the Lie algebra characteristic to classical mechanics.

Definition 1.7.26 (Poisson Algebra). *A Poisson algebra is an associative commutative algebra $\mathfrak{P}$ equipped with a Lie bracket $\{\cdot, \cdot\}$ acting as a derivation with respect to the associative product. Namely it satisfies the Leibniz rule*

$$\{f, lh\} = \{f, l\}h + l\{f, h\} \quad \forall f, h, l \in \mathfrak{P}.$$

In order to reconstruct classical mechanics in an algebraic way, the structure defined in Definition 1.7.26 must be given and must be a subalgebra of $\mathfrak{A}$ as defined in Definition 1.7.14. In addition, to recover the smooth structure of the phase space, we need an isomorphism to a particular $C_0^\infty(M)$. These strong requirements are coded inside the following definition.

Definition 1.7.27 (Smooth Dense Poisson Subalgebra). *Let $\mathfrak{A}$ be a commutative unital C^* -algebra as in Definition 1.7.10 with Gelfand spectrum X as in Definition 1.7.21. A smooth dense Poisson subalgebra is a Poisson subalgebra $(\mathfrak{A}^\infty, \{\cdot, \cdot\})$ with $\mathfrak{A}^\infty \subset \mathfrak{A}$ satisfying*

1. $\mathfrak{A}^\infty$ is a selfadjoint subset as in Definition 1.7.3,
2. $\mathfrak{A}^\infty$ is dense in $\mathfrak{A}$ with respect to the uniform norm,
3. there exists a smooth manifold M such that $\mathfrak{A}^\infty \cong C_0^\infty(M)$ as algebras,
4. $\mathfrak{A}^\infty$ is closed under the Poisson bracket $\{\cdot, \cdot\}$.

We can now ask if the Poisson and the smooth structure can be extended to the whole space of characters X . The first answer is negative due to the behaviour of the Poisson brackets presented in Definition 1.7.26.

Remark 1.7.28 (Topological Obstruction to Poisson Structure). The Poisson bracket, as seen in Definition 1.7.26, acts as a first-order differential operator. This makes it intrinsically unbounded and discontinuous with respect to the uniform supremum norm of the C^* -algebra $\mathfrak{A}$. This obstruction necessitates restricting the Poisson structure strictly to the incomplete dense subalgebra $\mathfrak{A}^\infty$, sacrificing Banach completeness to preserve the Leibniz rule.

On the other hand, investigating the smooth property, we can find that there is a technical way to extend that structure. In fact, if M is compact, we can prove the following result.

Proposition 1.7.29. *Let $\mathfrak{A}$ be a commutative unital C^* -algebra as in Definition 1.7.10 with Gelfand spectrum X as in Definition 1.7.21. Given $(\mathfrak{A}^\infty, \{\cdot, \cdot\})$ the smooth dense Poisson subalgebra, as in Definition 1.7.27, with $\mathfrak{A}^\infty \cong C_0^\infty(M)$. If M is compact, X is a smooth manifold.*

Proof. Since M is compact, $C_0^\infty(M) = C^\infty(M)$ contains the constant functions, so via the isomorphism of Definition 1.7.27 we have $1_{\mathfrak{A}} \in \mathfrak{A}^\infty$. In particular $\mathfrak{A}^\infty$ is a unital subalgebra of $\mathfrak{A}$.

Define the evaluation map

$$\text{ev} : X \rightarrow \sigma(\mathfrak{A}^\infty), \quad \text{ev}(x) = \text{ev}_x := x|_{\mathfrak{A}^\infty}, \quad \text{ev}_x(f) = f(x) \quad \forall f \in \mathfrak{A}^\infty,$$

which is well defined and continuous, being dual to the inclusion $\mathfrak{A}^\infty \hookrightarrow \mathfrak{A}$.

Injectivity. If $x_1|_{\mathfrak{A}^\infty} = x_2|_{\mathfrak{A}^\infty}$, then x_1 and x_2 agree on the dense subset $\mathfrak{A}^\infty$ (Definition 1.7.27) and are both norm-continuous, so $x_1 = x_2$ on $\mathfrak{A}$.

Surjectivity. Under $\mathfrak{A}^\infty \cong C_0^\infty(M)$, every character of $\mathfrak{A}^\infty$ is evaluation at some $p \in M$, and its value is bounded by the sup norm on M , namely by the $\mathfrak{A}$ -norm restricted to $\mathfrak{A}^\infty$. Being a bounded multiplicative functional on the dense subalgebra $\mathfrak{A}^\infty \subset \mathfrak{A}$, it extends uniquely to a bounded multiplicative functional on $\mathfrak{A}$, becoming an element of X . Hence ev is surjective.

Thus ev is a continuous bijection. Since $\mathfrak{A}$ is unital, X is compact; since $\sigma(\mathfrak{A}^\infty) \cong M$ is Hausdorff, a continuous bijection from a compact space to a Hausdorff space is automatically closed, hence a homeomorphism. Composing with $\sigma(\mathfrak{A}^\infty) \cong M$ gives a homeomorphism $X \cong M$.

Transporting the smooth atlas of M along this homeomorphism equips X with a smooth structure, making it a smooth manifold diffeomorphic to M . $\square$

Remark 1.7.30. The hypothesis M compact is equivalent to $1_{\mathfrak{A}} \in \mathfrak{A}^\infty$, and is generally easy to check directly on the manifold underlying $\mathfrak{A}^\infty$ — e.g. for the qubit, $M = S^2$ is compact, so the hypothesis holds automatically.

After all this discussion, we can collect the algebraic and differential ingredients identified so far into a single, self-contained structure.

Definition 1.7.31 (Classical Algebraic System). *A classical mechanical system is completely characterized by the triplet $(\mathfrak{A}, \mathfrak{A}^\infty, \{\cdot, \cdot\})$, where:*

1. $\mathfrak{A} \cong C^0(X)$ is a commutative unital C^* -algebra, as in Definition 1.7.11, determining the topological phase space X , its Gelfand spectrum.
2. $\mathfrak{A}^\infty \subset \mathfrak{A}$ is a smooth dense Poisson subalgebra, as in Definition 1.7.27, inducing the differential structure.
3. $\{\cdot, \cdot\}$ is a Poisson bracket, as in Definition ??, defined exclusively on $\mathfrak{A}^\infty$.

The last step is to show that this is sufficient to recover the complete symplectic manifold. This can be done invoking the representation theory of Poisson algebras. Rather than imposing the symplectic structure externally, one can use Landsman's representation theorem [5, Theorem 2.6.7]. It establishes that the irreducible representations of this triplet are classified by the symplectic manifolds obtained from the restriction of the Poisson bracket on submanifolds of the Poisson structure. This recovers the full phase space construction seen before.

The representation theory of Poisson algebras has been explained further in [5, Section 2.6].

Physical Observables and Dynamics

Before ending this chapter, we must clarify the physical interpretation of the algebraic structures introduced so far. We previously established the classical Gelfand algebra $\mathfrak{A} \cong C^0(X)$ in Definition 1.7.13 to reconstruct the topological space, and we isolated the measurable physical quantities within its selfadjoint sector $\mathfrak{A}_{sa}$ in Definition 1.7.14. However, to formulate Hamiltonian mechanics, we also required the smooth dense Poisson subalgebra $\mathfrak{A}^\infty$ from Definition 1.7.27.

We can assemble these topological, algebraic, and differential requirements in the following way:

- The space of all continuous measurable observables is the selfadjoint sector of the full classical Gelfand algebra $\mathfrak{A}$, namely $\mathfrak{A}_{sa} = \{f \in \mathfrak{A} \mid f^* = f\}$. This space is sufficient to define states, probability measures, and continuous symmetries, but structurally inadequate to generate differential dynamics since it lacks a Poisson bracket.
- Hamiltonian mechanics requires smoothness to formulate the equations of motion. Consequently, the Hamiltonian function H must belong to the smooth selfadjoint sector, defined as $\mathfrak{A}_{sa}^\infty := \mathfrak{A}^\infty \cap \mathfrak{A}_{sa}$. This exact requirement guarantees that the Hamiltonian vector field $X_H = \{H, \cdot\}$ generates a well-defined, strictly real one-parameter group of symplectomorphisms $\alpha_t : \mathfrak{A}^\infty \rightarrow \mathfrak{A}^\infty$.

A non-selfadjoint Hamiltonian, or one lacking smoothness, would either produce complex flows or exit the domain of the Poisson bracket, yielding unphysical dynamics. Thus, restricting the dynamical generators to the selfadjoint part of $\mathfrak{A}^\infty$ ensures the simultaneous preservation of the real structure of physical observables and the symplectic geometry of the underlying phase space.

1.8 Example I: Classical Spin System

In this section, we apply the algebraic tools developed in the first chapter to the classical spin system. This example provides an important test case for our framework, since it is a non-trivial Poisson algebra whose reduction yields a compact curved manifold, the sphere, allowing every abstract construction introduced so far, from the Gelfand spectrum to the Poisson bracket, to be verified explicitly on a concrete and geometrically intuitive example. Working through this case in detail also illustrates how symmetries of a physical system translate, via Noether's theorem, into conserved quantities within the same algebraic language.

1.8.1 The Poisson Algebra and Gelfand's Theorem

Consider the polynomial algebra $\mathfrak{A}$ generated by three elements x_1, x_2, x_3 , equipped with the $\mathfrak{so}(3)$ -Poisson bracket

$$\{x_i, x_j\} = \epsilon_{ijk} x_k, \quad i, j, k \in \{1, 2, 3\},$$

where ϵ_{ijk} is the Levi-Civita symbol. This bracket is skew-symmetric and satisfies the Jacobi identity by virtue of the Lie algebra structure. A direct computation reveals the existence of a Casimir function

$$C = x_1^2 + x_2^2 + x_3^2, \quad \{C, A\} = 0 \quad \forall A \in \mathfrak{A}.$$

Since $\mathfrak{A}$ is purely abstract, we must first extract a space from it. Fix $C = 1$ by identifying $A \sim A'$ whenever $A - A' = Q(C - 1)$ for some $Q \in \mathfrak{A}$. Call $\mathfrak{A}_1$ the resulting quotient. This identification respects the bracket precisely because C is a Casimir, that is, if $A - A' = Q(C - 1)$, then $\{A, B\} - \{A', B\} = (C - 1)\{Q, B\}$, which again vanishes in the quotient. So $\mathfrak{A}_1$ inherits a well-defined Poisson bracket.

The points of $\mathfrak{A}_1$ are the multiplicative linear functionals $\omega : \mathfrak{A}_1 \rightarrow \mathbb{R}$ in the dual space $\mathfrak{A}_1^*$. We call X their collection. Since x_1, x_2, x_3 generate $\mathfrak{A}_1$, each $\omega \in X$ is fixed by the triple $(\omega(x_1), \omega(x_2), \omega(x_3)) \in \mathbb{R}^3$. Multiplicativity and $x_1^2 + x_2^2 + x_3^2 = 1$ in $\mathfrak{A}_1$ force this triple onto $\mathbb{S}^2$; conversely every point of $\mathbb{S}^2$ defines such a functional by evaluation. Hence

$$X \cong \mathbb{S}^2,$$

obtained here as a consequence of the algebra, not assumed from the start. Completing $\mathfrak{A}_1$ in the uniform norm on X gives a commutative C^* -algebra isomorphic to $C^0(\mathbb{S}^2)$, consistent with Gelfand's Theorem 1.1, and the bracket descends with it, making $\mathbb{S}^2$ a natural candidate for a symplectic manifold.

1.8.2 Construction of the Symplectic Leaf

We now equip the leaf with coordinates and explicitly construct the symplectic form. Use standard spherical coordinates

$$x_1 = \sin \theta \cos \phi, \quad x_2 = \sin \theta \sin \phi, \quad x_3 = \cos \theta,$$

where $\theta \in [0, \pi]$ and $\phi \in [0, 2\pi)$. Now we must compute the bracket $\{\theta, \phi\}$. From $\{x_1, x_2\} = x_3 = \cos \theta$, we have

$$\{x_1, x_2\} = \frac{\partial x_1}{\partial \theta} \frac{\partial x_2}{\partial \phi} \{\theta, \phi\} + \frac{\partial x_1}{\partial \phi} \frac{\partial x_2}{\partial \theta} \{\phi, \theta\}.$$

Substituting the partial derivatives yields

$$\{x_1, x_2\} = \sin \theta \cos \theta \{\theta, \phi\}.$$

We equate this to $x_3 = \cos \theta$ getting

$$\boxed{\{\theta, \phi\} = \frac{1}{\sin \theta}}. \quad (1.3)$$

This formula is valid away from the poles. Its extension to $\sin \theta = 0$ is given by the Cartesian expression, ensuring global non-degeneracy.

Inversion to the Symplectic Form

For a two-dimensional manifold, we can invert the commutator found in Equation (1.3) to obtain the symplectic form

$$\boxed{\omega = \sin \theta \, d\theta \wedge d\phi}. \quad (1.4)$$

Equivalently, in Cartesian coordinates restricted to $\mathbb{S}^2$, this is the standard area form

$$\omega = x_1 dx_2 \wedge dx_3 + x_2 dx_3 \wedge dx_1 + x_3 dx_1 \wedge dx_2 \Big|_{\mathbb{S}^2}.$$

This form is closed ($d\omega = 0$) and non-degenerate, thus $(\mathbb{S}^2, \omega)$ is a symplectic manifold.

1.8.3 Hamiltonian Action and the Moment Map

The group $G = SO(3)$ acts on $\mathbb{S}^2$ by rotations. For each $\xi \in \mathfrak{so}(3) \cong \mathbb{R}^3$, the infinitesimal generator is given by

$$X_\xi(x) = x \times \xi,$$

where $\times$ denotes the standard cross product. We seek a moment map $\mu : \mathbb{S}^2 \rightarrow \mathfrak{so}(3)^*$ satisfying the fundamental relation

$$d\langle \mu(x), \xi \rangle = \iota_{X_\xi} \omega, \quad \forall \xi \in \mathfrak{so}(3).$$

Identifying $\mathfrak{so}(3)^* \cong \mathbb{R}^3$ via the dot product, we test the candidate $\mu(x) = -x$. Take $\xi = e_1 = (1, 0, 0)$. Then $X_{e_1} = (0, x_3, -x_2)$. Contracting ω with X_{e_1}

$$\begin{aligned} \iota_{X_{e_1}} \omega &= \iota_{(0, x_3, -x_2)} (x_1 dx_2 \wedge dx_3 + x_2 dx_3 \wedge dx_1 + x_3 dx_1 \wedge dx_2) \\ &= x_1(x_3 dx_3 + x_2 dx_2) - (x_2^2 + x_3^2) dx_1. \end{aligned}$$

Using the constraint $x_2 dx_2 + x_3 dx_3 = -x_1 dx_1$, we get

$$\iota_{X_{e_1}} \omega = -dx_1.$$

Thus, $d\langle \mu, e_1 \rangle = d(-x_1)$, which implies

$$\boxed{\mu(x) = -x}. \quad (1.5)$$

Physically, this is the negative of the position vector, which corresponds to the angular momentum vector, up to a sign convention.

1.8.4 Noether's Theorem in Practice

The moment map constructed above is the precise algebraic counterpart of a conserved quantity associated with a continuous symmetry, in the sense of Noether's theorem. We now illustrate this correspondence concretely through two physically motivated examples, the precession of a classical system in a magnetic field and the symmetric spinning top, both governed by the same underlying symplectic structure on the sphere.

Example 1: Precession in a Magnetic Field

Consider the Hamiltonian

$$H = x_3,$$

which is invariant under rotations about the z -axis. The symmetry generator is $\xi = e_3 = (0, 0, 1)$. The associated Noether charge is

$$\langle \mu, \xi \rangle = \langle -x, e_3 \rangle = -x_3.$$

as calculated in Equation (1.5) Noether's theorem guarantees x_3 is conserved. The equations of motion are

$$\begin{aligned}\dot{x}_1 &= \{x_1, H\} = -x_2, \\ \dot{x}_2 &= \{x_2, H\} = x_1, \\ \dot{x}_3 &= \{x_3, H\} = 0.\end{aligned}$$

Solving yields a uniform precession around the z -axis:

$$x_1(t) + ix_2(t) = e^{it}(x_1(0) + ix_2(0)), \quad x_3(t) = x_3(0).$$

Example 2: The Symmetric Rigid Body

Now take the Hamiltonian

$$H = \frac{1}{2}(x_1^2 + x_2^2) = \frac{1}{2}(1 - x_3^2).$$

This is also invariant under the z -axis rotations, so the conserved charge remains $-x_3$. However, the dynamics become richer. The equations of motion are

$$\begin{aligned}\dot{x}_1 &= \{x_1, H\} = x_2 x_3, \\ \dot{x}_2 &= \{x_2, H\} = -x_1 x_3, \\ \dot{x}_3 &= 0.\end{aligned}$$

The solution is again a rotation about the z -axis, but with an angular velocity that depends on the conserved quantity:

$$\dot{\phi} = -x_3 = \text{const.}$$

This matches the behavior of a symmetric Euler top.